

- The integral of a function of the density

$$\mathcal{F}(\mu) := \begin{cases} \int_{\Omega} f(\rho(x)) dx & \text{if } \mu = \rho \cdot dx, \\ +\infty & \text{otherwise.} \end{cases}$$

- The sum of a function of the masses of the atoms

$$\mathcal{G}(\mu) := \begin{cases} \sum_i g(a_i) & \text{if } \mu = \sum_i a_i \delta_{x_i}, \\ +\infty & \text{if } \mu \text{ is not purely atomic.} \end{cases}$$

The scope of this chapter is to study some properties of these functionals. The first questions that we analyze are classical variational issues (semi-continuity, convexity, first variation...). Then, we also introduce and analyze a new notion, the notion of geodesic convexity. This is a natural concept in the analysis on metric spaces: is a certain functional convex *along geodesics* of the metric space? In the case of the Wasserstein spaces, and in particular of $\mathbb{W}_2$, this concept, introduced by McCann in [229], is called *displacement convexity*. It turns out to be quite different than usual convexity, but very useful in many cases.

7.1 semi-continuity

Since the main goal is to use these functionals in variational models, in order to be able to minimize them, we first give some semi-continuity criteria. We will consider semi-continuity or continuity of these functionals with respect to weak convergence (meaning as usual the duality with C_b functions). We start from the most simple functionals.

7.1.1 Potential & interaction energies, transport costs, dual norms

We start this review from the easiest case: the potential energy.

Proposition 7.1. *If $V \in C_b(\Omega)$ then $\mathcal{V}$ is continuous for the weak convergence of probability measures. If V is l.s.c. and bounded from below then $\mathcal{V}$ is semi-continuous. Moreover, semi-continuity of V (respectively, continuity) is necessary for the semi-continuity (continuity) of $\mathcal{V}$.*

Proof. The continuity of $\mu \mapsto \int V d\mu$ for $V \in C_b$ is straightforward by definition of weak convergence. If V is l.s.c. and bounded from below, we know that there is a sequence of Lipschitz and bounded functions V_k increasingly converging to V (see Box 1.5). Then, by monotone convergence, we infer that $\mathcal{V}(\mu) = \lim_k \int V_k d\mu = \sup_k \int V_k d\mu$. This allows to see $\mathcal{V}$ as a supremum of continuous functionals, and hence it is l.s.c.

The necessity part is straightforward if one considers sequences of points $x_k \rightarrow x$ and the associated Dirac masses. Since $\delta_{x_k} \rightarrow \delta_x$ then continuity of $\mathcal{V}$ implies $V(x_k) \rightarrow V(x)$ and semi-continuity implies $\liminf_k V(x_k) \geq V(x)$. $\square$ $\square$

We pass now to the case of the interaction functional $\mathcal{W}$.

Proposition 7.2. *If $W \in C_b(\Omega)$ then $\mathcal{W}$ is continuous for the weak convergence of probability measures. If W is l.s.c. and bounded from below then $\mathcal{W}$ is semi-continuous.*

This result is an easy consequence of the following lemma (that we prove for subsets of $\mathbb{R}^d$ but could be adapted to general locally compact spaces).

Lemma 7.3. *Let Ω be a closed subset of $\mathbb{R}^d$, $\mu_k, \mu \in \mathcal{P}(\Omega)$ and $\mu_k \rightharpoonup \mu$ then*

$$\mu_k \otimes \mu_k \rightharpoonup \mu \otimes \mu$$

as probabilities on $\Omega \times \Omega$.

Proof. We want to prove that for any function $\phi \in C_b(\Omega \times \Omega)$ we have the convergence $\int \phi d\mu_k \otimes \mu_k \rightarrow \int \phi d\mu \otimes \mu$. First consider the case $\phi(x, y) = \chi(x)\zeta(y)$. In this case

$$\int_{\Omega \times \Omega} \phi d\mu_k \otimes \mu_k = \int_{\Omega} \chi d\mu_k \cdot \int_{\Omega} \zeta d\mu_k \rightarrow \int_{\Omega} \chi d\mu \cdot \int_{\Omega} \zeta d\mu = \int_{\Omega \times \Omega} \phi d\mu \otimes \mu.$$

This proves that the desired convergence is true for all functions $\phi \in \mathcal{A}(\Omega)$, where the class $\mathcal{A}(\Omega)$ is given by

$$\mathcal{A}(\Omega) = \left\{ \phi(x, y) = \sum_{i=1}^N \chi_i(x) \zeta_i(y), \quad N \in \mathbb{N}, \chi_i, \zeta_i \in C_b(\Omega) \right\}.$$

The class $\mathcal{A}$ is an algebra of functions which contains the constants and separates the points of $\Omega \times \Omega$. Hence, by Stone-Weierstrass theorem (see Box 2.3 in Section 3.1), if $\Omega \times \Omega$ is compact (i.e. if Ω is compact), it is dense in $C(\Omega \times \Omega)$. It is a general fact that weak-* convergence in a dual space may be tested on a dense set, provided that the sequence we are considering is bounded. Here weak convergence corresponds in the compact case to the weak-* convergence in the duality with $C(\Omega \times \Omega)$ and we are considering a sequence of probability measures, which is bounded. This proves $\mu_k \otimes \mu_k \rightharpoonup \mu \otimes \mu$ in the compact case.

If Ω is not compact, one can use Stone-Weierstrass theorem to prove that the convergence holds against any compactly supported function. The conclusion follows if we recall that a sequence of probability measures weakly converges (in the duality with C_b) if and only if it weakly-* converges in the duality with C_c . This was proven in Lemma 5.8. $\square$

It is worthwhile to note that besides these two classes of functionals one could consider higher order interaction energies defined through

$$\mu \mapsto \int_{\Omega} \int_{\Omega} \dots \int_{\Omega} W(x_1, x_2, \dots, x_n) d\mu(x_1) d\mu(x_2) \dots d\mu(x_n).$$

In particular the set of all these functional could be considered as “polynomials” on the space $\mathcal{P}(\Omega)$ and some analysis of the space of functions over measures are based on this very class of functionals¹.

We pass now to some different functionals.

¹This is an approach used by P.-L. Lions in his lectures at Collège de France, see [212].

Proposition 7.4. *For any $p < +\infty$, the Wasserstein distance $W_p(\cdot, \nu)$ to any fixed measure $\nu \in \mathcal{P}(\Omega)$ is continuous w.r.t. weak convergence provided Ω is compact. If Ω is not compact and $\nu \in \mathcal{P}_p(\Omega)$, then $W_p(\cdot, \nu)$ is well-defined and finite-valued over $\mathcal{P}_p(\Omega)$ and it is only l.s.c.*

More generally, for any continuous cost $c : \Omega \times \Omega \rightarrow \mathbb{R}$ the functional $\mathcal{T}_c(\cdot, \nu)$ is continuous if Ω is compact. Without compactness on Ω , if c is l.s.c. and bounded from below, then $\mathcal{T}_c(\cdot, \nu)$ is l.s.c.

Proof. We start from the compact case. For Wasserstein distances, continuity is straightforward since the convergence for the distance W_p exactly metrizes the weak convergence and every distance is continuous w.r.t. itself (as a consequence of triangle inequality). For general transport costs $\mathcal{T}_c$, just apply Theorem 1.51.

For the non-compact case, we give a unified proof. Consider a sequence $\mu_k \rightharpoonup \mu$ and a sequence of optimal transport plans $\gamma_k \in \Pi(\mu_k, \nu)$ for a cost c , l.s.c. and bounded from below (which includes the case $c(x, y) = |x - y|^p$). Since μ_k is tight, γ_k is also tight. First extract a subsequence such that $\lim_h \mathcal{T}_c(\mu_{k_h}, \nu) = \liminf_k \mathcal{T}_c(\mu_k, \nu)$ and then extract once more so as to guarantee $\gamma_{k_h} \rightharpoonup \gamma$ for some γ . We know that image measures through continuous functions pass to the limit, so that we obtain $\gamma \in \Pi(\mu, \nu)$. Hence,

$$\mathcal{T}_c(\mu, \nu) \leq \int c(x, y) d\gamma \leq \liminf_h \int c(x, y) d\gamma_{k_h} = \lim_h \mathcal{T}_c(\mu_{k_h}, \nu) = \liminf_k \mathcal{T}_c(\mu_k, \nu).$$

Here the first inequality comes from the fact that γ is admissible but maybe not optimal. The second from semi-continuity of the integral of c (Proposition 7.1). $\square$ $\square$

Finally, here is another class of functionals with some interesting examples.

Proposition 7.5. *Let $\mathcal{X}$ be a Banach space of functions over Ω such that $\mathcal{X} \cap C_b(\Omega)$ is dense in $\mathcal{X}$. Then*

$$\mu \mapsto \|\mu\|_{\mathcal{X}'} := \sup_{\varphi \in \mathcal{X}, \|\varphi\| \leq 1} \int \varphi d\mu = \sup_{\varphi \in \mathcal{X} \setminus \{0\}} \frac{\int \varphi d\mu}{\|\varphi\|}$$

(a functional which is set to $+\infty$ if $\mu \notin \mathcal{X}'$) is l.s.c. for the weak convergence.

Proof. This fact is straightforward if one notes that we can write

$$\|\mu\|_{\mathcal{X}'} = \sup_{\varphi \in C_b(\Omega) \cap \mathcal{X}, \|\varphi\| \leq 1} \int \varphi d\mu,$$

which expresses $\|\mu\|_{\mathcal{X}'}$ as a supremum of linear functionals, continuous for the weak convergence. $\square$ $\square$

It is interesting to see two examples.

Proposition 7.6. *Suppose $\mathcal{X} = H_0^1(\Omega)$ (where $\Omega \subset \mathbb{R}^d$ is a smooth compact domain, and we endow this space with the L^2 norm of the gradient) and let, for every $\mu \in H^{-1}(\Omega)$, φ_μ be the solution of*

$$\begin{cases} -\Delta \varphi = \mu & \text{in } \Omega, \\ \varphi = 0 & \text{on } \partial\Omega. \end{cases}$$

Then $\|\mu\|_{H^{-1}}^2 = \int \varphi_\mu d\mu = \iint G(x, y) d\mu(x) d\mu(y)$, where $G(x, \cdot)$ is the Green function of the Dirichlet Laplacian, defined as the solution of

$$\begin{cases} -\Delta\varphi = \delta_x & \text{in } \Omega, \\ \varphi = 0 & \text{on } \partial\Omega. \end{cases}$$

In particular, if $\Omega = (-1, 1) \subset \mathbb{R}$, then

$$G(x, y) = \frac{1}{2}((1+x) \wedge (1+y))((1-x) \wedge (1-y)).$$

Suppose, instead, $\mathcal{X} = H^1(\Omega)$ (endowed with the full H^1 norm) and let, for every $\mu \in (H^1(\Omega))' = \mathcal{X}'$, φ_μ be the solution of

$$\begin{cases} -\Delta\varphi + \varphi = \mu & \text{in } \Omega, \\ \frac{\partial\varphi}{\partial n} = 0 & \text{on } \partial\Omega. \end{cases}$$

Then $\|\mu\|_{\mathcal{X}'}^2 = \int \varphi_\mu d\mu = \iint G(x, y) d\mu(x) d\mu(y)$, where $G(x, \cdot)$ is defined as the solution of

$$\begin{cases} -\Delta\varphi + \varphi = \delta_x & \text{in } \Omega, \\ \frac{\partial\varphi}{\partial n} = 0 & \text{on } \partial\Omega. \end{cases}$$

In particular, if $\Omega = (-1, 1) \subset \mathbb{R}$, then

$$G(x, y) = \cosh((1+x) \wedge (1+y)) \cosh((1-x) \wedge (1-y)) / \sinh(2).$$

Proof. We first prove that $\|\mu\|_{\mathcal{X}'}^2 = \int \varphi_\mu d\mu$ in the two cases.

First case : H_0^1 . Note that, for every function $\psi \in H_0^1$, one has

$$\int \psi d\mu = \int \nabla\psi \cdot \nabla\varphi_\mu \leq \|\psi\|_{\mathcal{X}} \|\varphi_\mu\|_{\mathcal{X}}$$

and equality holds for $\psi = \varphi_\mu$ (we use the homogeneous norm $\|\psi\|_{\mathcal{X}} := \|\nabla\psi\|_{L^2}$). This shows that the supremum is realized by a multiple of φ_μ and that $\int \varphi_\mu d\mu = \|\varphi_\mu\|_{\mathcal{X}}^2$. As a consequence, we have

$$\|\mu\|_{\mathcal{X}'}^2 = \left(\frac{\int \varphi_\mu d\mu}{\|\varphi_\mu\|_{\mathcal{X}}} \right)^2 = \int \varphi_\mu d\mu.$$

The case of H^1 is similar, with the only exception of the first computation

$$\int \psi d\mu = \int \nabla\psi \cdot \nabla\varphi_\mu + \psi\varphi_\mu \leq \|\psi\|_{\mathcal{X}} \|\varphi_\mu\|_{\mathcal{X}}$$

and, again, equality holds for $\psi = \varphi_\mu$.

Finally, expressing $\int \varphi_\mu d\mu$ as a double integral $\iint G(x, y) d\mu(x) d\mu(y)$ is only a matter of expressing $\varphi_\mu(x)$ as $\int G(x, y) d\mu(y)$. This is possible by using the theory of Green functions and, for the one dimensional case $\Omega = (-1, 1)$, it is enough to compute that $\int G(x, y) d\mu(y)$ is a solution of the desired equation. $\square$ $\square$

In this way we have seen that, thanks to Green functions, we have expressed these dual norms functionals as interaction functionals.

7.1.2 Local Functionals

Local functionals over measures are defined as those functionals $F : \mathcal{M}(\Omega) \rightarrow \mathbb{R}$ such that $F(\mu + \nu) = F(\mu) + F(\nu)$ whenever μ and ν are mutually singular (i.e. there exists $A, B \subset \Omega$ with $A \cup B = \Omega$, $\mu(A) = 0$ and $\nu(B) = 0$). This obviously does not make much sense for probability measures, since, if μ and ν are probabilities, then $\mu + \nu$ is not a probability. Yet, we will see in this section some functionals that can be defined over probability measures but could be also seen as the restriction to probabilities of functionals defined on more general measures. All our proofs could be easily generalized to finite positive measures; on the contrary, for simplicity we will not consider at all the case of signed or vector measures. We refer to [68] for the general theory of these functionals.

First, let us consider functionals of the form

$$\mu = \rho \cdot \lambda \mapsto \int f(\rho(x)) \lambda(dx),$$

where λ is a given positive measure over Ω (for instance the Lebesgue measure).

Let us first see which are the natural conditions on f so as to ensure lower semi-continuity. Then, we will give a precise result.

As a first point, consider a sequence ρ_n of densities, taking values a and b on two sets A_n and B_n chosen in the following way: we fix a partition of Ω into small cubes of diameter $\varepsilon_n \rightarrow 0$ and then we select set A_n and B_n such that, for every cube Q of this partition, we have $\lambda(A_n \cap Q) = (1-t)\lambda(Q)$ and $\lambda(B_n \cap Q) = t\lambda(Q)$. We can check that ρ_n weakly converges to a uniform density $\rho = (1-t)a + tb$. semi-continuity of the functional would imply

$$f((1-t)a + tb) \leq (1-t)f(a) + tf(b),$$

i.e. convexity of f . Hence it is clear that one needs to require f to be convex.

Another requirement concerns the growth of f . For simplicity it is always possible to assume $f(0) = 0$ (up to adding a constant to f). Suppose that f satisfies $f(t) \leq Ct$ for all $t \geq 0$. Consider the functional defined through

$$\mathcal{F}(\mu) : \begin{cases} \int f(\rho(x)) d\lambda(x) & \text{if } \mu = \rho \cdot \lambda, \\ +\infty & \text{otherwise.} \end{cases}$$

Take a sequence μ_n of absolutely continuous probability measures weakly converging to a singular measure μ : we get $\mathcal{F}(\mu_n) \leq C$ while $\mathcal{F}(\mu) = +\infty$, thus violating the semi-continuity. This suggests that one should have $C = +\infty$, i.e. f superlinear. The following theorem gives a general result which also includes compensation for non-superlinearity.

Proposition 7.7. *Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ be a convex l.s.c. function, and set $L := \lim_{t \rightarrow \infty} f(t)/t = \sup_{t > 0} f(t)/t \in \mathbb{R} \cup \{+\infty\}$. Let λ be a fixed finite positive measure on Ω . For every measure μ write $\mu = \rho \cdot \lambda + \mu^s$, where $\rho \cdot \lambda$ is the absolutely continuous part of μ and μ^s be the singular one (w.r.t. λ). Then, the functional defined through*

$$\mathcal{F}(\mu) = \int_{\Omega} f(\rho(x)) d\lambda(x) + L\mu^s(\Omega)$$

(note that, if $L = +\infty$, then $F(\mu) = +\infty$ whenever μ has a singular part) is l.s.c.

Proof. We will try to use the equality

$$f(t) = \sup_{a \in \mathbb{R}} at - f^*(a).$$

This equality is a consequence of $f^{**} = f$, which is itself due to the fact f is convex and lower semi-continuous. Moreover, it is easy to see that for $a > L$ we have $f^*(a) = +\infty$: indeed, $f^*(a) = \sup_t at - f(t) \geq \lim_{t \rightarrow +\infty} t(a - f(t)/t)$. Hence, we can write

$$f(t) = \sup_{a \leq L} at - f^*(a) = \sup_{a < L} at - f^*(a)$$

(the last equality is justified by the fact that f^* is continuous on the set where it is finite).

Let us consider the following functional

$$\widetilde{\mathcal{F}}(\mu) := \sup_{a \in C_b(\Omega), \sup a < L} \int a(x) d\mu(x) - \int f^*(a(x)) d\lambda(x).$$

$\widetilde{\mathcal{F}}$ is obviously l.s.c. since it is the supremum of a family of affine functionals, continuous w.r.t. the weak convergence. We want to prove $\widetilde{\mathcal{F}} = \mathcal{F}$.

In order to do so, first note that, thanks to Lusin's theorem (see Box 1.6 in Section 1.1), applied here to the measure $\lambda + \mu$, it is not difficult to replace bounded and continuous functions with measurable bounded functions. By abuse of notation, we denote measurable bounded functions by $L^\infty(\Omega)$ (even if we do not mean functions which are essentially bounded w.r.t. a given measure, but really bounded) and we get

$$\widetilde{\mathcal{F}}(\mu) := \sup_{a \in L^\infty(\Omega), \sup a < L} \int a(x) d\mu(x) - \int f^*(a(x)) d\lambda(x).$$

Then take a set A such that $\mu^s(\Omega \setminus A) = \lambda(A) = 0$: this allows to write

$$\widetilde{\mathcal{F}}(\mu) := \sup_{a \in L^\infty(\Omega), \sup a < L} \int_{\Omega \setminus A} [a(x)\rho(x) - f^*(a(x))] d\lambda(x) + \int_A a(x) d\mu^s(x).$$

The values of $a(x)$ may be chosen independently on A and $\Omega \setminus A$ and we can check that

$$\begin{aligned} \sup_{a \in L^\infty(\Omega), \sup a < L} \int_{\Omega \setminus A} [a(x)\rho(x) - f^*(a(x))] d\lambda(x) &= \int f(\rho(x)) d\lambda(x), \\ \sup_{a \in L^\infty(\Omega), \sup a < L} \int_A a(x) d\mu^s(x) &= L\mu^s(A) = L\mu^s(\Omega), \end{aligned}$$

which allows to conclude $\widetilde{\mathcal{F}} = \mathcal{F}$. □ □

Remark 7.8. The assumption that λ is a finite measure is necessary to avoid integrability issues for the term $\int f^*(a(x)) d\lambda$. A typical case where this term could give troubles is the entropy $f(t) = t \log t$, where $f^*(s) = e^{s-1}$. Suppose that λ is the Lebesgue

measure on the whole $\mathbb{R}^d$; it is easy to see that, for any L^∞ function a , the integral $\int_{\mathbb{R}^d} e^{a(x)-1} dx$ does not converge, and this provides $\widehat{\mathcal{F}}(\mu) = -\infty$ for every μ . However, if λ is σ -finite (which is the case of the Lebesgue measure on $\mathbb{R}^d$) and $f \geq 0$, it is possible to prove the semi-continuity by approximating λ from below with finite measures².

Remark 7.9. The reader could be shocked by the fact that in Proposition 7.7 we did not require lower bounds on f . In particular, the same result is true for $f(t) = t$ and $f(t) = -t$, which implies continuity of the mass with respect to the weak convergence of measures! Indeed, the weak convergence that we are using is the one in duality with C_b , and it is perfectly normal that the mass (i.e. the integral against the function 1) is continuous in this convergence. If we used, instead, the weak-* convergence in duality with C_c on a non-compact space, the situation would be different. In order to perform the same proof, one would need at least $L > 0$ (otherwise, the functions $a \in C_c$ do not satisfy $a > L$), or $L = 0$ and $f^*(0) < +\infty$.

If we come back to the interpretation of $\mathcal{F}$, it is not difficult to check that $\mathcal{F}$ “favors” dispersed measure: first it is only finite for absolutely continuous measures, second, due to the Jensen inequality, the value of $\mathcal{F}$ is minimal for the constant density.

If we look for functionals having an opposite behavior and favoring the concentrated part of the measure, there are at least two different choices. We can look at an interaction functional such as $\mu \mapsto \int \int |x - y|^2 d\mu(x) d\mu(y)$ (where the square of the distance could be replaced by any increasing function of it). This is a global and spatially dependent functional, and has a different flavour than $\mathcal{F}$. Indeed, we can find in the same class of local functionals some functionals which favor concentration, by looking in particular at the atomic part of μ . It is the case of the functional

$$\mathcal{G}(\mu) := \begin{cases} \sum_i g(a_i) & \text{if } \mu = \sum_i a_i \delta_{x_i}, x_i \neq x_j \text{ for } i \neq j, \\ +\infty & \text{if } \mu \text{ is not purely atomic.} \end{cases}$$

As before, let us first understand which are the basic properties of g so as to guarantee semi-continuity.

We also assume $g(0) = 0$ (which is by the way necessary if we want to avoid ambiguities due to zero-mass atoms). Suppose that g satisfies $g(t) \leq Ct$ for all $t > 0$ and take, similar to what we did before, a sequence $\mu_n \rightarrow \mu$ where μ_n is purely atomic but μ is not (take for instance the sequence in Proposition 4.38). Then we have $\mathcal{G}(\mu_n) \leq C$ but $\mathcal{G}(\mu) = +\infty$. This is a contradiction with semi-continuity and suggests that we should consider functions g such that $\lim_{t \rightarrow 0} g(t)/t = +\infty$.

Second consideration, take $\mu_n = a\delta_{x_n} + b\delta_{y_n} + \nu$ (with ν finitely atomic: we only use ν so as to come back to a total mass of 1). Suppose $x_n \rightarrow x$ and $y_n \rightarrow x$ and $x \notin \text{spt}(\nu)$. Then $\mu_n \rightarrow \mu = (a+b)\delta_x + \nu$. In this case semi-continuity would imply

$$g(a+b) \leq g(a) + g(b),$$

i.e. we need g to be subadditive.

It is useful to establish the following.

²For the case of the entropy, the lower semi-continuity on the whole $\mathbb{R}^d$ is false, but it is true under stronger convergence assumptions, see Ex(45). This is the usual strategy to study on the whole space the variational problems that we will present in bounded domains in Section 8.3 for the Fokker-Planck equation.

Lemma 7.10. *Suppose that $g(0) = 0$ and that g is subadditive. Suppose that $\mu_n = \sum_{i=1}^N a_{i,n} \delta_{x_{i,n}}$ is a sequence of atomic probability measures with bounded number of atoms. Then, up to a subsequence, we have $\mu_n \rightharpoonup \mu$ where μ is also atomic with at most N atoms and $\mathcal{G}(\mu) \leq \liminf \mathcal{G}(\mu_n)$.*

Proof. Let us precise that we impose to the points $x_{i,n}$ to be distinct. If a measure μ_n has less than N atoms then we chose points $x_{i,n}$ at random to complete the list of atoms and set $a_{i,n} = 0$ for those extra indices.

Now extract a subsequence (not relabeled) such that for each $i = 1, \dots, N$ one has $a_{i,n} \rightarrow a_i$ and $x_{i,n} \rightarrow x_i$. For this subsequence one has $\mu_n \rightharpoonup \mu := \sum_{i=1}^N a_i \delta_{x_i}$. It is possible that the points x_i are not distinct. If they are distinct we have $\mathcal{G}(\mu) = \sum_{i=1}^N g(a_i)$, otherwise we have (thanks to subadditivity) $\mathcal{G}(\mu) \leq \sum_{i=1}^N g(a_i)$. Anyway we have

$$\mathcal{G}(\mu) \leq \sum_{i=1}^N g(a_i) = \lim_n \sum_{i=1}^N g(a_{i,n}) = \lim_n \mathcal{G}(\mu_n),$$

which proves the desired result (one can choose the subsequence so that it realizes the liminf of the whole sequence). $\square$ $\square$

It is now possible to prove the following

Lemma 7.11. *Suppose that $g(0) = 0$, that $g(t) \geq 0$, that g is subadditive and l.s.c. and that $\lim_{t \rightarrow 0} g(t)/t = +\infty$. Then $\mathcal{G}$ is l.s.c. on $\mathcal{P}(\Omega)$.*

Proof. Assume without loss of generality that all the μ_n are purely atomic. Fix a number $M > 0$ and use $\lim_{t \rightarrow 0} g(t)/t = +\infty$: this implies that there exists ε_0 such that for all $t < \varepsilon_0$ we have $g(t) > Mt$. Consider a sequence $\mu_n \rightharpoonup \mu$, assume $\mathcal{G}(\mu_n) \leq C < +\infty$ and decompose it into $\mu_n = \mu_n^s + \mu_n^b$, where $\mu_n^b = \sum_{i=1}^N a_{i,n} \delta_{x_{i,n}}$ is the sum of the atoms of μ_n with mass at least ε_0 . In particular, these atoms are no more than $N := \varepsilon_0^{-1}$. The other part μ_n^s (the “small” part, μ_n^b being the “big” one) is just defined as the remaining atoms (every μ_n is purely atomic since $\mathcal{G}(\mu_n) < +\infty$).

If we write

$$C \geq \mathcal{G}(\mu_n) \geq \mathcal{G}(\mu_n^s) = \sum_i g(a_{i,n}) \geq M \sum_i a_{i,n} = M \mu_n^s(\Omega)$$

we get an estimate on the mass of the “small” part. Hence it is possible to get, up to subsequences,

$$\mu_n^b \rightharpoonup \mu^b \quad \text{and} \quad \mu_n^s \rightharpoonup \mu^s, \quad \mu^s(\Omega) \leq \frac{C}{M}.$$

We can now apply lemma 7.10 to prove that μ^b is purely atomic and that $\mathcal{G}(\mu^b) \leq \liminf_n \mathcal{G}(\mu_n^b) \leq \liminf_n \mathcal{G}(\mu_n)$.

This proves that μ must be purely atomic, since the possible atomless part of μ must be contained in μ^s , but $\mu^s(\Omega) \leq C/M$. Hence the mass of the non-atomic part of μ must be smaller than C/M for every $M > 0$, i.e. it must be zero.

We have now proven that μ is purely atomic and we have an estimate of $\mathcal{G}(\mu^b)$, where μ^b is a part of μ depending on M . If we write $(a_i)_i$ for the masses of μ and $(a_i^M)_i$ for those of μ^b we have

$$\sum_i g(a_i^M) \leq \liminf_n \mathcal{G}(\mu_n) := \ell.$$

We want to prove $\sum_i g(a_i) \leq \ell$ and, to this aim, it is enough to let $M \rightarrow \infty$. Actually, $\mu^s(\Omega) = \sum_i (a_i - a_i^M) \leq C/M \rightarrow 0$ implies that for each i we have $a_i - a_i^M \rightarrow 0$ and thus $a_i^M \rightarrow a_i$. Using the semi-continuity of g we have $g(a_i) \leq \liminf_{M \rightarrow \infty} g(a_i^M)$. If we fix an arbitrary number N we get

$$\sum_{i=1}^N g(a_i) \leq \liminf_{M \rightarrow \infty} \sum_{i=1}^N g(a_i^M) \leq \ell.$$

By passing to the supremum over N we finally get

$$\sum_{i=1}^{\infty} g(a_i) \leq \ell,$$

which is the claim. $\square$

As a particular example for the functionals of type $\mathcal{F}$ we can consider the L^p norms to the power p , i.e.

$$\mathcal{F}(\mu) = \|\mu\|_{L^p}^p = \begin{cases} \int \rho(x)^p dx & \text{if } \mu = \rho \cdot dx, \\ +\infty & \text{otherwise.} \end{cases}$$

For the functionals of type $\mathcal{G}$, we can instead consider the cardinality of the support, obtained for $g(t) = 1$ if $t > 0$ and $g(0) = 0$:

$$\mathcal{G}(\mu) = \#(\text{spt}(\mu)).$$

Box 7.1. – Good to know! – Characterization of local l.s.c. functionals on $\mathcal{M}^d(\Omega)$

It is also useful to note that functionals of the form $\mathcal{F}$ and $\mathcal{G}$ could be mixed, obtaining local functionals accepting both absolutely continuous and atomic parts. There is a general lower semi-continuity result that comes from the general theory developed in [68, 69, 70], which also characterizes the semi-continuity. It also covers the case of vector-valued measures and can be expressed as follows.

Theorem - Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex and l.s.c. and $g : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be subadditive and l.s.c. For every vector $\mathbf{v} \in \mathbb{R}^d$ define

$$f^\infty(\mathbf{v}) := \lim_{t \rightarrow +\infty} \frac{f(t\mathbf{v})}{t} \quad \text{and} \quad g^0(\mathbf{v}) := \lim_{t \rightarrow 0} \frac{g(t\mathbf{v})}{t},$$

and suppose $f^\infty = g^0$. Given a finite positive measure λ , decompose every vector measure $\mu \in \mathcal{M}^d(\Omega)$ as $\mu = \rho \cdot \lambda + \mu^c + \mu^\#$, where $\mu^c + \mu^\#$ is the singular part of μ w.r.t. λ , which is also decomposed into a purely atomic part $\mu^\# = \sum_i a_i \delta_{x_i}$ and the remainder, usually referred to as the Cantor part; $\rho : \Omega \rightarrow \mathbb{R}^d$ is a vector density and the $a_i = \mu(\{x_i\}) \in \mathbb{R}^d$ are also vectors; we also denote by $\mathbf{w} : \Omega \rightarrow S^{d-1}$ the density of μ^c w.r.t. its own total variation measure $|\mu^c|$. Then the functional

$$\mu \mapsto \int f(\rho(x)) d\lambda(x) + \int f^\infty(\mathbf{w}(x)) d|\mu^c|(x) + \sum_i g(a_i)$$

is local and lower semi-continuous for the weak convergence of measures.

Conversely, every local and lower semi-continuous functional on $\mathcal{M}^d(\Omega)$ can be written in the form above, for suitable choices of f , g and λ .

7.2 Convexity, first variations and subdifferentials

We pass in this section to another important notion concerning these functionals. If lower semi-continuity is crucial to establish existence results for minimization problems, convexity properties (in particular, strict convexity) are crucial for uniqueness. Also, with convexity comes the notion of subdifferential, thus we come to another very natural question in calculus of variations: how to compute first variations of these functionals?

We start from the very first question: which among these functionals are convex?

Convexity and strict convexity

- $\mathcal{V}$ is linear and hence convex but not strictly convex.
- $\mathcal{W}$ is quadratic, but not always convex. Take for instance $W(x, y) = |x - y|^2$ and compute

$$\begin{aligned} & \int_{\Omega} \int_{\Omega} |x - y|^2 d\mu(x) d\mu(y) \\ &= \int_{\Omega} \int_{\Omega} |x|^2 d\mu(x) d\mu(y) + \int_{\Omega} \int_{\Omega} |y|^2 d\mu(x) d\mu(y) - 2 \int_{\Omega} \int_{\Omega} x \cdot y d\mu(x) d\mu(y) \\ &= 2 \int_{\Omega} |x|^2 d\mu(x) - 2 \left(\int_{\Omega} x d\mu(x) \right)^2. \end{aligned}$$

This shows that $\mathcal{W}$ has (in this case) a linear part to which we subtract the square of another linear one; it is then concave rather than convex.

- All transport costs can be expressed by duality formula as a supremum of linear functionals

$$\mathcal{T}_c(\mu, \nu) = \sup_{\varphi(x) + \psi(y) \leq c(x, y)} \int \varphi d\mu + \int \psi d\nu$$

and it is hence convex (but W_p is in general not). Strict convexity is discussed later (it is true for $c(x, y) = |x - y|^p$, $p > 1$ and $\nu \ll \mathcal{L}^d$).

- The norm in a dual functional space is always convex since it is a norm, but is never strictly convex because it is 1-homogeneous. Note that also the square of a norm could be non-strictly convex (as it is the case for the L^∞ or the L^1 norms).
- The functionals $\mathcal{F}$ that we considered above are actually convex due to the assumptions on f . Strict convexity is true if f is strictly convex and $L = +\infty$ (for instance if one takes $f(t) = \sqrt{1 + t^2} - 1$, which is strictly convex, then $\mathcal{F}$ is not strictly convex because it is finite and linear on singular measures).
- On the contrary, the functionals $\mathcal{G}$ that we considered above are typically not convex since the typical examples of sub-additive functions are concave.